

Lab 3 - Routing Protocol

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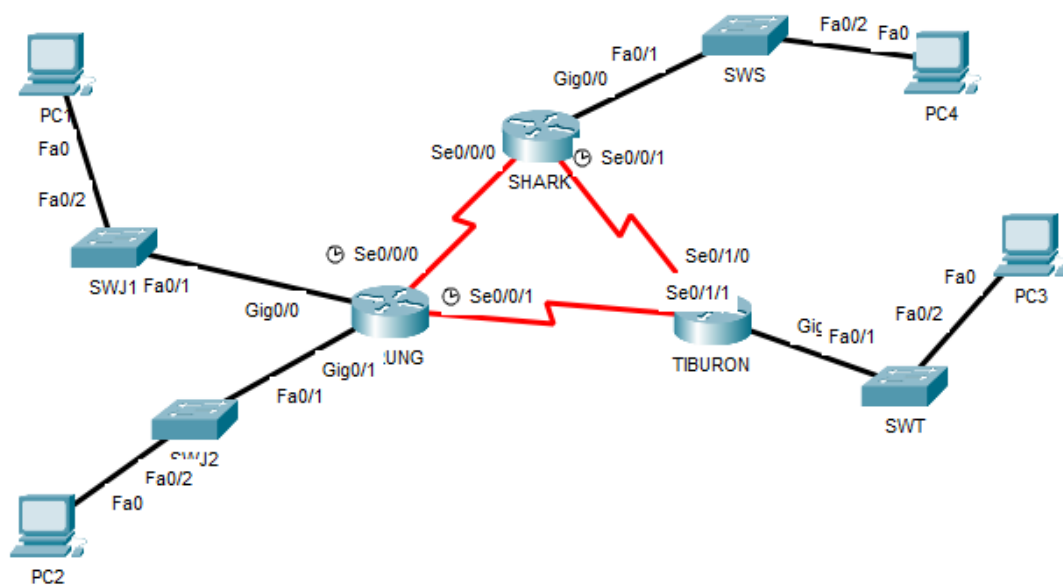
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SECTION: 12

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.1	255.255.255.224	-
4		G0/1	172.18.110.33	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.129	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.65	255.255.255.192	-
11	PC1	-	172.18.110.30	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

- Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.

- JERUNG LAN1 (20 hosts): /27 (30 usable IPs) → **172.18.110.0/27 (172.18.110.1 – 172.18.110.30)**
 $2^5 = 32$
 Subnet mask: /27
 Usable IPs: $32 - 2 = 30$
- JERUNG LAN2 (30 hosts): /27 (30 usable IPs) → **172.18.110.32/27 (172.18.110.33 - 172.18.110.62)**
 $2^5 = 32$
 Subnet mask: /27
 Usable IPs: $32 - 2 = 30$
- SHARK LAN (60 hosts): /26 (62 usable IPs) → **172.18.110.64/26 (172.18.110.65 – 172.18.110.126)**
 $2^6 = 64$
 Subnet mask: /26
 Usable IPs: $64 - 2 = 62$
- TIBURON LAN (50 hosts): /26 (62 usable IPs) → **172.18.110.128/26 (172.18.110.129 - 172.18.110.190)**
 $2^6 = 64$
 Subnet mask: /26
 Usable IPs: $64 - 2 = 62$
- Connections: Point-to-point links need only 2 IPs each (/30)
 - JERUNG-SHARK (**172.18.110.193 - 172.18.110.194**)
 - JERUNG-TIBURON (**172.18.110.197 - 172.18.110.198**)
 - SHARK-TIBURON (**172.18.110.201-172.18.110.202**) $2^2 = 4$
 Subnet mask: /30
 Usable IPs: $4 - 2 = 2$

Subnet Mask in binary and decimal form

Device	Subnetwork	Prefix	Subnet Mask
JERUNG	LAN1	/27	11111111.11111111.11111111.11100000 = 255.255.255.224
	LAN2	/27	11111111.11111111.11111111.11100000 = 255.255.255.224
SHARK	LAN	/26	11111111.11111111.11111111.11000000 = 255.255.255.192
TIBURON	LAN	/26	11111111.11111111.11111111.11000000 = 255.255.255.192
Connections	JERUNG-SHARK	/30	11111111.11111111.11111111.11111100 = 255.255.255.252
	JERUNG-TIBURON	/30	11111111.11111111.11111111.11111100 = 255.255.255.252
	SHARK-TIBURON	/30	11111111.11111111.11111111.11111100 = 255.255.255.252

- Using the IP addresses calculated, configure the devices with the appropriate information.

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.

- a. CLI

- i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

- b. PT tool

- i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

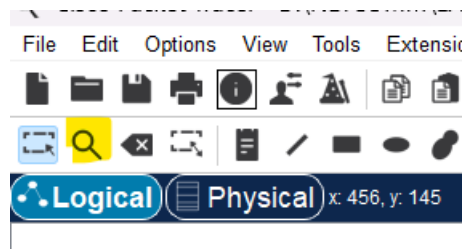


Figure B

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

JERUNG:

CLI

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
 i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
 * - candidate default, U - per-user static route, o - ODR
 P - periodic downloaded static route

Gateway of last resort is not set

```

    172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.33/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

```

PT tool

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0

SHARK:

CLI

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
 i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
 * - candidate default, U - per-user static route, o - ODR
 P - periodic downloaded static route

Gateway of last resort is not set

```

    172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

```

PT tool

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0

TIBURON:

[CLI](#)

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
 i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
 * - candidate default, U - per-user static route, o - ODR
 P - periodic downloaded static route

Gateway of last resort is not set

```

      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.128/26 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0

```

[PT tool](#)

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time=3ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

3. Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. Explain the reason(s) behind the results.

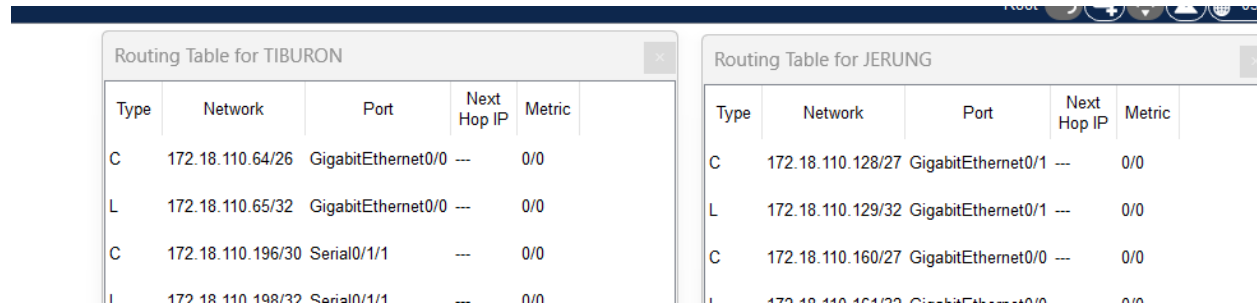
This is because PC1 and PC4 are on different subnet, causing them unable to communicate directly to each other. Besides that, they are connected to different routers. Because of that, the result shows “Destination host unreachable”.

5. What needs to be done to ensure all PCs can ping each other successfully?

- All interfaces must have the proper IP addresses and subnet masks.
- Default gateways must be configured on all PCs.
- Verify that no access control lists (ACLs) are blocking communication.
- RIP or static routes must be configured so that routing information can be shared between routers.

Task 3 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.



Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

- In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- What can you say about the addresses used in the 'network' instructions in Figure E?

The 'network' instructions are used to tell RIP to advertise and include all interface within the subnets.

network 172.18.110.128 – include all interface within 172.18.110.128 subnet

network 172.18.110.160 – include all interface within 172.18.110.160 subnet

network 172.18.110.192 – include all interface within 172.18.110.192 subnet

network 172.18.110.196 – include all interface within 172.18.110.196 subnet

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0 ---		0/0
L	172.18.110.1/32	GigabitEthernet0/0 ---		0/0
C	172.18.110.32/27	GigabitEthernet0/1 ---		0/0
L	172.18.110.33/32	GigabitEthernet0/1 ---		0/0
R	172.18.110.64/26	Serial0/0/0	172.18.110.194	120/1
R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.192/30	Serial0/0/0 ---		0/0
L	172.18.110.193/32	Serial0/0/0 ---		0/0
C	172.18.110.196/30	Serial0/0/1 ---		0/0
L	172.18.110.197/32	Serial0/0/1 ---		0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1
R	172.18.110.200/30	Serial0/0/0	172.18.110.194	120/1

TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.128/26	GigabitEthernet0/0 ---		0/0
L	172.18.110.129/32	GigabitEthernet0/0 ---		0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1 ---		0/0
L	172.18.110.198/32	Serial0/1/1 ---		0/0
C	172.18.110.200/30	Serial0/1/0 ---		0/0
L	172.18.110.202/32	Serial0/1/0 ---		0/0

- a. What are the changes seen in TIBURON?

Before RIP, the routing table only showed **directly connected networks**. After RIP, TIBURON's table includes **RIP-learned routes (R)** for the networks connected to JERUNG and SHARK.

- b. What are the Networks with type R in TIBURON and JERUNG?

TIBURON:

- 172.18.110.0/27 (LAN1 on JERUNG)
- 172.18.110.32/27 (LAN2 on JERUNG)
- 172.18.110.64/26 (SHARK LAN)

JERUNG:

- 172.18.110.64/26 (SHARK LAN)
- 172.18.110.128/26 (TIBURON LAN)

- c. Ping PC3 from PC1. Was it successful?

Yes

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.190: bytes=32 time=7ms TTL=126
Reply from 172.18.110.190: bytes=32 time=5ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 4ms
```

- d. Ping PC4 from PC1. Was it successful?

Yes

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.126: bytes=32 time=13ms TTL=126
Reply from 172.18.110.126: bytes=32 time=23ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 23ms, Average = 12ms
```

- e. Explain the reasons for your answer.

This is because RIP allows routers to dynamically exchange routing information, enabling end-to-end communication between PCs in different subnets. Besides that, correct configurations of IPs, subnet masks, and default gateways also ensure the packets are forwarded to their destinations.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.64
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
```

- g. Open router SHARK's routing table and paste here.

```
R    172.18.110.0/27   Serial0/0/0         172.18.110.193 120/1

R    172.18.110.32/27 Serial0/0/0         172.18.110.193 120/1

C    172.18.110.64/26 GigabitEthernet0/0 ---          0/0

L    172.18.110.65/32 GigabitEthernet0/0 ---          0/0

R    172.18.110.128/26 Serial0/0/1         172.18.110.202 120/1

C    172.18.110.192/30 Serial0/0/0         ---          0/0

L    172.18.110.194/32 Serial0/0/0         ---          0/0

R    172.18.110.196/30 Serial0/0/0         172.18.110.193 120/1

R    172.18.110.196/30 Serial0/0/1         172.18.110.202 120/1

C    172.18.110.200/30 Serial0/0/1         ---          0/0

L    172.18.110.201/32 Serial0/0/1         ---          0/0
```

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	Success
2	Success
3	Success

PC1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=14ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=2ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 4ms

C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=9ms TTL=126
Reply from 172.18.110.30: bytes=32 time=17ms TTL=126
Reply from 172.18.110.30: bytes=32 time=7ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 17ms, Average = 8ms
```

PC2

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=7ms TTL=126
Reply from 172.18.110.62: bytes=32 time=6ms TTL=126
Reply from 172.18.110.62: bytes=32 time=7ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 5ms

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=12ms TTL=126
Reply from 172.18.110.62: bytes=32 time=2ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 4ms
```

PC3

```
C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=7ms TTL=126
Reply from 172.18.110.190: bytes=32 time=12ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 5ms

C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=10ms TTL=126
Reply from 172.18.110.190: bytes=32 time=5ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 4ms
```

Task 4 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: [192.168.1.0/24](#)
 - b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.2
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?
No

Ping PC3 from PC4

```
Pinging 192.168.1.2 with 32 bytes of data:

Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Ping PC3 from PC1

```
Pinging 198.162.1.2 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Request timed out.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 198.162.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

SHARK

R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

TIBURON

R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0
C	192.168.1.0/26	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

- e. Referring to the routing table, explain your findings.
- In routing table for JERUNG, contains RIP-learned (R) routes for networks such as 172.18.110.0/27, 172.18.110.32/27, and 172.18.110.64/26, which are from **TIBURON** or **SHARK**.
 - TIBURON's routing table contains RIP-learned (R) routes for networks 172.18.110.0/27, 172.18.110.32/27, and 172.18.110.64/26, which are part of **JERUNG** and **SHARK**'s LANs.
 - From both routing tables, we can prove that both **JERUNG** and **TIBURON** rely on RIP.
- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?
- Ensure all routers have the correct network statements in RIP to propagate 192.168.1.0/24. This can be done by updating RIP on **TIBURON** and propagate the new route for other subnets.
- g. Show your configurations in TIBURON to ensure end-to-end connectivity.
- ```
TIBURON>enable
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#no auto-summary
```
- The network 192.168.1.0 command tells the router to advertise and learn routes in this subnet range.
- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

| PC | Ping result |
|----|-------------|
| 1  | Success     |
| 2  | Success     |
| 4  | Success     |

## PC1

```
C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=13ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.30:
 Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
 Minimum = 1ms, Maximum = 13ms, Average = 5ms

C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.30:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
 Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

## PC2

```
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.62:
 Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
 Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.62:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
 Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

## PC4

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.126: bytes=32 time=2ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.126:
 Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=2ms TTL=126
Reply from 172.18.110.126: bytes=32 time=2ms TTL=126

Ping statistics for 172.18.110.126:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
 Approximate round trip times in milli-seconds:
 Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

## ***REFLECTION***

What have you learned in this task?

From lab 3, I have obtained many useful knowledge to improve my understanding about routing protocol. For example, subnetting helps to improve efficiency of IP allocation. Besides that, RIP simplifies routing process between routers but requires proper configuration to prevent error.